Statistical Generative Modeling

Lecture 1: What does it mean to generate data?

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From prediction to generation

Classical ML: prediction

Use observed input x to estimate an unknown target y:

$$p_\theta(y \mid x).$$

- regression
- classification

Modern GenAI: generation

Produce realistic samples X from a complex data distribution (using a prompt or context c):

$$X \sim p_\theta(x) \quad \text{or} \quad X \sim p_\theta(x \mid c).$$

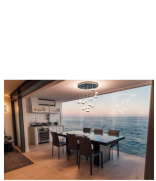
- images, text, audio, and video
- code and molecular structures

Richard Feynman: "What I cannot create, I do not understand"

Generative modeling: "What I understand, I can create"

Discriminative vs. generative

Discriminative: classify bedroom vs. dining room






The image X is given. **Goal**: decision boundary, via **conditional distribution over label Y**

P(Y = Bedroom | X= )

Generative: generate X

Y=B , X=  Y=D , X= 

Y=B , X=  Y=D , X= 

... ...

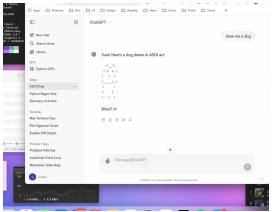
Y=B , X=  Y=D , X= 

The input X is **not** given. Requires a model of the **joint distribution over both X and Y**

P(Y = Bedroom , X= )

Illustration from the supplied CS236 reference deck (Ermon, 2023).

Generation across modalities



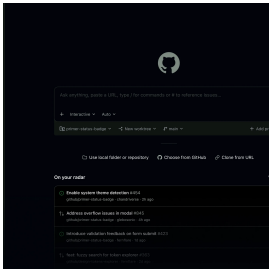
Language
ChatGPT



Images
GPT Image



Video
Veo

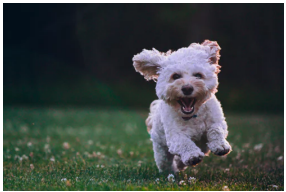


Code
GitHub Copilot

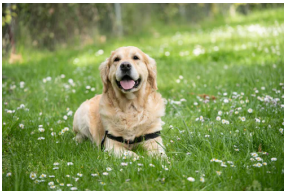
Generative models output complex objects, not just numbers.

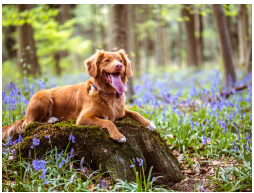
Official public examples from OpenAI, Google DeepMind, and GitHub.

Realism and diversity









A useful generator must produce samples that are

- **realistic**,
- **diverse**, and
- appropriate for the requested context.

Generation is a distributional problem, not a point-prediction problem.

Images from the supplied CS236 reference deck (Ermon, 2023).

What is a generative model?

Image: $X \in \mathbb{R}^{H \times W \times 3}$,

Video: $X \in \mathbb{R}^{T \times H \times W \times 3}$,

Text: $X = (X_1, \ldots, X_T)$.

We imagine an unknown population distribution

$$X \sim p_{\text{data}}(x),$$

but only observe a finite dataset

$$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} p_{\text{data}}.$$

A generative model aims to approximate p_{data} well enough to produce realistic samples.

Learning to sample

Choose a model family $\{p_\theta : \theta \in \Theta\}$ and fit it to the observed data:

$$\widehat{\theta} \leftarrow X_{1:n} \qquad \text{so that} \qquad p_{\widehat{\theta}}(x) \approx p_{\text{data}}(x).$$

Then generate a new observation by sampling

$$X_{\text{new}} \sim p_{\widehat{\theta}}(x).$$

Two key tasks:

Learning

Compare the model with the data and estimate θ.

Sampling

Use the fitted model to produce new random draws.

Conceptual framing follows the supplied Fundamentals and CS236 reference lectures.

Conditional generation

Unconditional generation

$$X \sim p_\theta(x).$$

Draw from the overall data distribution.

Conditional generation

$$X \sim p_\theta(x \mid C = c).$$

Draw from the distribution consistent with a label, prompt, partial observation, or scientific constraint.



Prompt: “An astronaut riding a horse”

Illustration from the supplied CS236 reference deck (Ermon, 2023).

Content generation



- **Language:** GPT, Claude, Gemini, Llama
- **Images:** GPT Image, Imagen, Stable Diffusion, Midjourney
- **Video:** Veo, Runway Gen-4.5, Kling
- **Code:** Codex, GitHub Copilot, Claude Code
- **Audio:** speech generators, Lyria, MusicGen

Conditions may be prompts, labels, reference images, partial drafts, or styles.

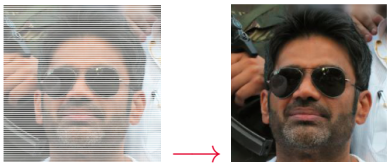
Across modalities, learn a distribution and sample from it.

Generated magazine example: OpenAI, ChatGPT Images 2.0.

Inverse problems



CT reconstruction



Masked image → completion

If Y is incomplete or noisy, use $p_\theta(x \mid y)$ to recover plausible values of X.

- **restoration:** denoise, deblur, colorize
- **completion:** inpaint, outpaint, impute
- **resolution:** recover detail from coarse inputs
- **scientific imaging:** sparse CT/MRI, microscopy
- **uncertainty:** generate several plausible solutions

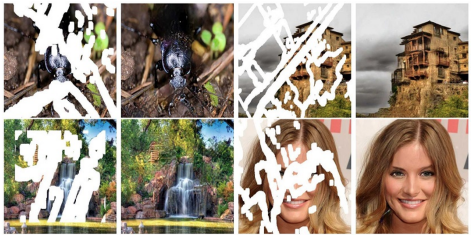
Progress in Inverse Problems

P(high resolution | low resolution)



Menon et al, 2020

P(full image| mask)



Liu al, 2018

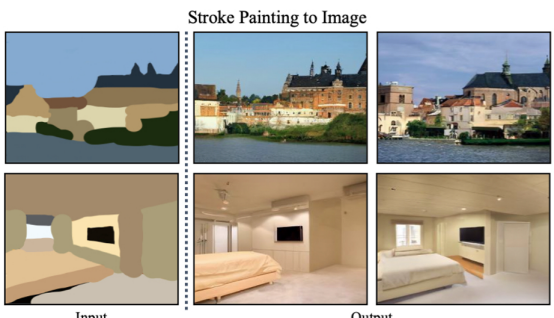
P(color image| greyscale)



Antic, 2020

Illustration from the supplied CS236 reference deck (Ermon, 2023).

Progress in Inverse Problems



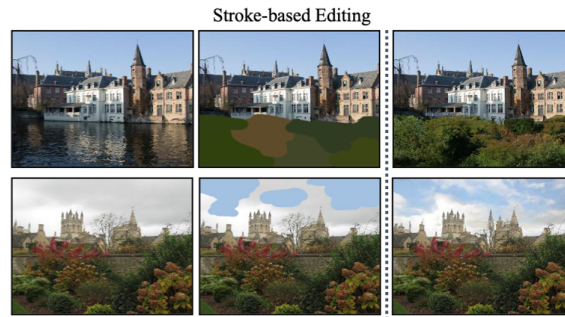


Illustration from the supplied CS236 reference deck (Ermon, 2023).

Progress in Inverse Problems

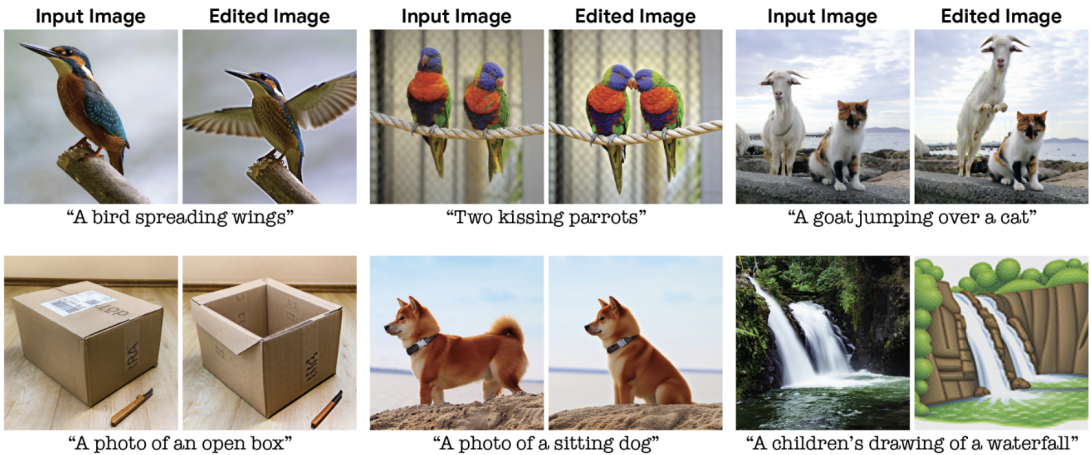
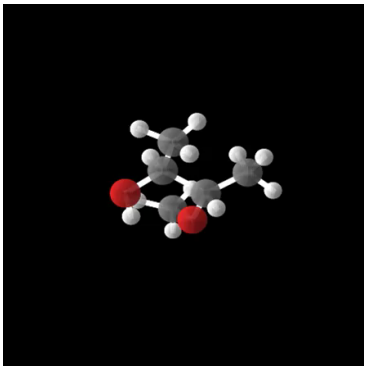


Illustration from the supplied CS236 reference deck (Ermon, 2023).

Scientific generation



- Molecules with desired chemical properties
- Proteins with structural or functional constraints
- Synthetic patient or omics data
- Realistic behavioral and time-series trajectories
- Simulation for uncertainty and sensitivity analysis

$$X \sim p_\theta(x \mid \text{desired properties}).$$

Molecule illustration from the supplied CS236 reference deck.

Why latent variables?

Introduce an unobserved variable Z and specify two simpler distributions:

$$\boxed{Z \sim p_\theta(z)} \qquad \Longrightarrow \qquad \boxed{X \mid Z = z \sim p_\theta(x \mid z)}.$$

The observed-data distribution is obtained by averaging over what we do not observe:

$$p_\theta(x) = \int p_\theta(x, z)\, dz = \int p_\theta(x \mid z) p_\theta(z)\, dz.$$

Z can encode a group, state, feature, source of variation, or intermediate stage of generation.

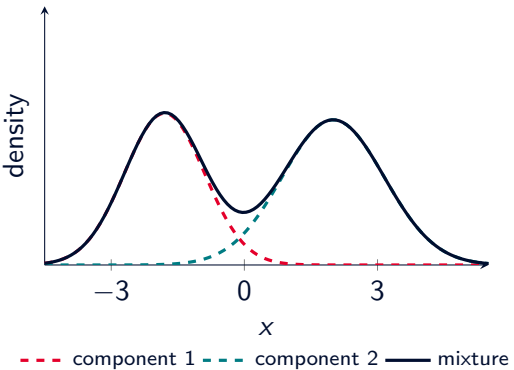
A mixture model

$$Z \sim \text{Categorical}(\pi_1, \pi_2),$$
$$X \mid Z = k \sim N(\mu_k, \sigma_k^2).$$

Marginalizing the unknown class gives

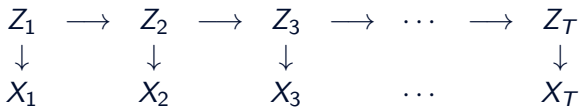
$$p(x) = \sum_{k=1}^{2} \pi_k \, \phi(x; \mu_k, \sigma_k^2).$$

Interpretation: Z selects a subpopulation before X is generated.



Latent states and dependence

For sequential data, let the hidden state evolve over time:



Generation

$$p(z_1) \prod_{t=2}^{T} p(z_t \mid z_{t-1}) \prod_{t=1}^{T} p(x_t \mid z_t).$$

Interpretation

- behavioral modes
- disease stages
- regimes or topics

Generation and inference

Generation: $Z \to X$

Draw $Z \sim p_\theta(z)$.

Draw $X \sim p_\theta(x \mid Z)$.

This is the model's forward story.

Inference: $X \to Z$

$$p_\theta(z \mid x) = \frac{p_\theta(x \mid z)p_\theta(z)}{p_\theta(x)}.$$

After observing x, infer which hidden explanation is plausible.

Generation runs the model forward; inference tries to invert it.

Why learning is hard

For observed data $x_{1:n}$, maximum likelihood requires

$$\widehat{\theta} \in \arg\max_{\theta} \sum_{i=1}^{n} \log p_\theta(x_i) = \arg\max_{\theta} \sum_{i=1}^{n} \log \int p_\theta(x_i, z_i)\, dz_i.$$

A sum or integral over hidden configurations may be computationally intractable.

Model structure	**Computational tool**
Mixtures	EM algorithm
Hidden Markov models	Forward–backward algorithm
Complex neural models	Variational inference and stochastic optimization

Three takeaways

A **generative model** describes a probability distribution or sampling mechanism for observed data.

Generative models support much more than content creation: they enable reconstruction, imputation, anomaly detection, scientific design, and uncertainty analysis.

A **latent variable** gives complex data a simpler hidden explanation—but introduces an inference problem.

Resources

Lectures

- Stanford CS236 (Deep Generative Models): `https://deepgenerativemodels.github.io/`
- UNC BIOS740 (Deep Learning Methods in Biomedical Sciences with PyTorch): `https://bios740.github.io/`

Textbooks

- Probabilistic Machine Learning (Advanced Topics) by Kevin Murphy: `https://probml.github.io/pml-book/book2.html`